CAS SciFinder Task History

Initiating Search

November 11, 2025, 5:14 PM

Search:

CAS Lexicon Search:

Sandmeyer reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (345)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (138) Filtered By: Yield: **90-100%, 80-89%, 70-79%** Document Type: **Journal**	**Reactions**	View Results

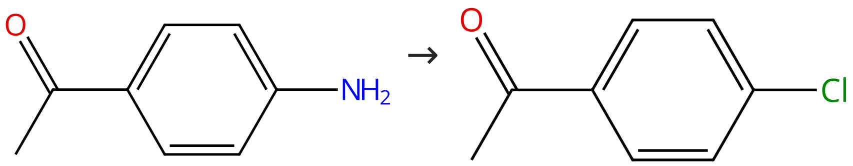
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (101)

Suppliers (67)

31-220-CAS-16613164

Steps: **1** Yield: **99%**

1.1 **Reagents:** *tert*-Butyl nitrite
Solvents: Acetonitrile; 1 min, rt

1.2 **Reagents:** Cupric chloride
Solvents: Acetonitrile, Ethylene glycol; 4 min, 82 °C

Experimental Protocols

Efficient Transposition of the Sandmeyer Reaction from Batch to Continuous Process

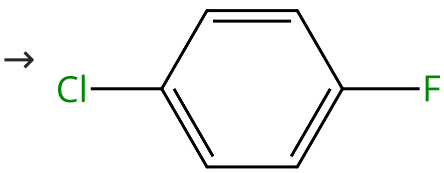
By: D'Attoma, Joseph; et al

Organic Process Research & Development (2017), 21(1), 44-51.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **98%**

Multi-component structure image available in CAS SciFinder



Suppliers (60)

Suppliers (30)

31-220-CAS-18799580

Steps: **1** Yield: **98%**

1.1 **Reagents:** Lithium chloride
Catalysts: Cuprous chloride, 1-Propanaminium, *N,N,N*-triethyl-, chloride (1:1) (silica supported); 1 h, 20 °C

Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

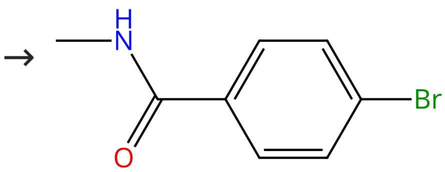
By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **97%**

Multi-component structure image available in CAS SciFinder



Suppliers (70)

31-614-CAS-46793030	Steps: **1** Yield: **97%**
1.1	**Reagents:** Acetonitrile, Bromotrichloromethane **Catalysts:** Barium titanate; 1 h, rt
1.2	**Solvents:** Ethyl acetate; rt
Experimental Protocols	

Piezoelectric-Driven Mechanochemical Halogenation: A Transformative Pathway for Sustainable Sandmeyer Reactions via Ball Milling

By: Gao, Pan; et al

Journal of Organic Chemistry (2025), 90(31), 11223-11229.

Scheme 4 (1 Reaction) Steps: **1** Yield: **97%**

Suppliers (78) Suppliers (59)

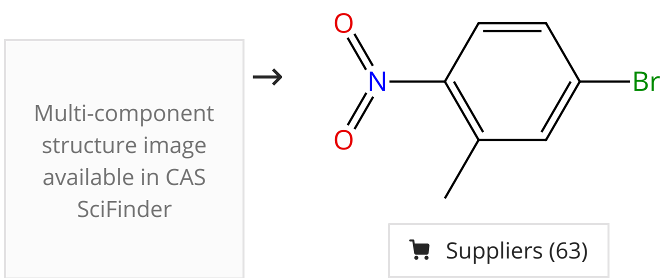
31-220-CAS-1185941	Steps: **1** Yield: **97%**
1.1	**Reagents:** Sulfuric acid **Solvents:** Water; rt → - 10 °C
1.2	**Reagents:** Sodium nitrite **Solvents:** Water; -10 - 0 °C; 2 h, - 5 - 0 °C; 0 °C → - 10 °C
1.3	**Reagents:** Hydrochloric acid, Cuprous chloride **Solvents:** Water; -15 - -10 °C; 3 h, rt

Synthesis of 2-chloro-5-iodobenzoic acid

By: Ye, Fei; et al

Yingyong Huagong (2011), 40(8), 1479-1480.

Scheme 5 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (14) Suppliers (63)

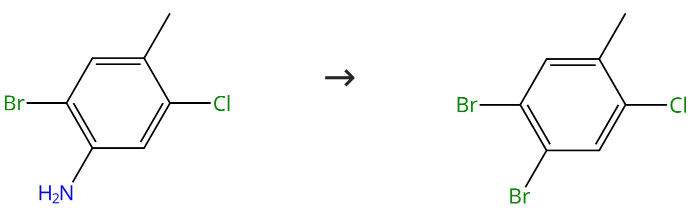
31-220-CAS-18799586	Steps: **1** Yield: **97%**
1.1	**Reagents:** Lithium bromide **Catalysts:** Cuprous chloride, 1-Propanaminium, *N,N,N*-triethyl-, chloride (1:1) (silica supported) **Solvents:** Acetonitrile; 5 min, rt
1.2	**Solvents:** Acetonitrile; 1 h, rt

Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

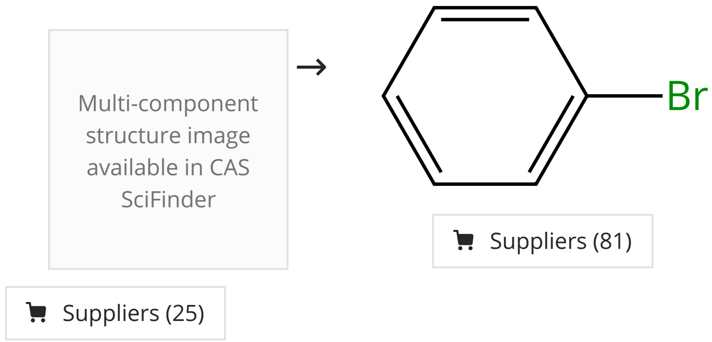
Scheme 6 (1 Reaction) Steps: **1** Yield: **96%**



Suppliers (64) Suppliers (44)

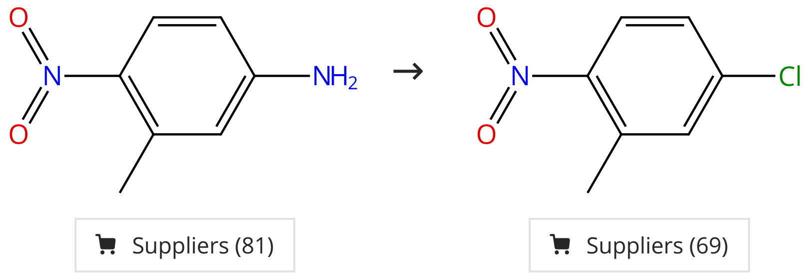
31-220-CAS-18799603	Steps: **1** Yield: **96%**	Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts
1.1	**Reagents:** Boron trifluoride etherate, Lithium bromide **Catalysts:** Cuprous chloride, 1-Propanaminium, *N,N,N*-triethyl-, chloride (1:1) (silica supported) **Solvents:** Acetonitrile; 10 min, 0 °C	By: Tarkhanova, Irina G.; et al Mendeleev Communications (2018), 28(3), 261-263.
1.2	**Reagents:** *tert*-Butyl nitrite **Solvents:** Acetonitrile; 1 h, 20 °C	

Scheme 7 (1 Reaction) Steps: **1** Yield: **96%**



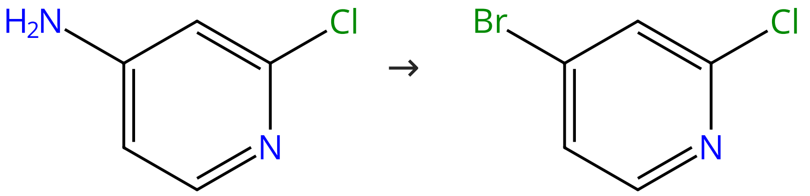
31-220-CAS-14140254	Steps: **1** Yield: **96%**	Cu(I)/Cu(II)/TMEDA, new effective available catalyst of Sandmeyer reaction
1.1	**Reagents:** Potassium bromide **Catalysts:** Copper bromide (CuBr), Copper bromide ($CuBr_2$), Dibenzo-18-crown-6 **Solvents:** Acetonitrile; 2 min	By: Sigeev, A. S.; et al Russian Journal of Organic Chemistry (2012), 48(8), 1055-1058.
1.2	**Catalysts:** *N,N,N',N'*-Tetramethylethylenediamine; 30 min, 20 °C	
1.3	**Solvents:** Acetonitrile; 20 °C; 1 h, 20 °C	
Experimental Protocols		

Scheme 8 (1 Reaction) Steps: **1** Yield: **96%**



31-220-CAS-18799596	Steps: **1** Yield: **96%**	Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts
1.1	**Reagents:** Tetraethylammonium chloride, Boron trifluoride etherate **Catalysts:** Cuprous chloride, 1-Propanaminium, *N,N,N*-triethyl-, chloride (1:1) (silica supported) **Solvents:** Acetonitrile; 10 min, 0 °C	By: Tarkhanova, Irina G.; et al Mendeleev Communications (2018), 28(3), 261-263.
1.2	**Reagents:** *tert*-Butyl nitrite **Solvents:** Acetonitrile; 1 h, 20 °C	

Scheme 9 (1 Reaction) Steps: **1** Yield: **96%**



Suppliers (106) Suppliers (106)

31-220-CAS-7547676 Steps: **1** Yield: **96%**

1.1 **Reagents:** *tert*-Butyl nitrite, Copper bromide ($CuBr_2$)
Solvents: Acetonitrile; rt; 15 min, rt; rt → 0 °C

1.2 **Solvents:** Acetonitrile; 0 °C; 1 h, 0 °C; 16 h, 0 °C → rt

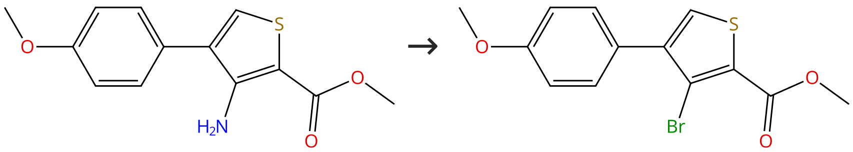
Experimental Protocols

Concise Entries to 4-Halo-2-pyridones and 3-Bromo-4-halo-2-pyridones

By: Honraedt, Aurelien; et al

Synlett (2016), 27(1), 67-69.

Scheme 10 (1 Reaction) Steps: **1** Yield: **95%**



Suppliers (26) Supplier (1)

31-220-CAS-1140956 Steps: **1** Yield: **95%**

1.1 **Reagents:** *tert*-Butyl nitrite, Copper bromide ($CuBr_2$)
Solvents: Acetonitrile; 70 °C; 4 h, 70 °C; 70 °C → rt

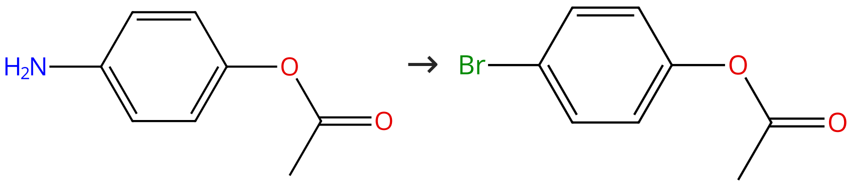
1.2 **Reagents:** Hydrochloric acid
Solvents: Water; rt

Novel and efficient access to thieno[3,4-c]cinnolines

By: Gehanne, Karine; et al

Heterocycles (2008), 75(12), 3015-3024.

Scheme 11 (1 Reaction) Steps: **1** Yield: **95%**



Suppliers (50) Suppliers (54)

31-220-CAS-18799599 Steps: **1** Yield: **95%**

1.1 **Reagents:** Boron trifluoride etherate, Lithium bromide
Catalysts: Cuprous chloride, 1-Propanaminium, *N,N,N*-triethyl-, chloride (1:1) (silica supported)
Solvents: Acetonitrile; 10 min, 0 °C

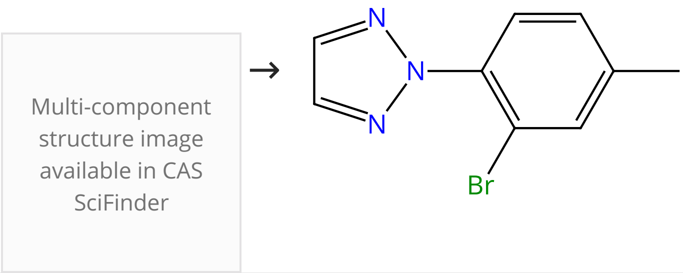
1.2 **Reagents:** *tert*-Butyl nitrite
Solvents: Acetonitrile; 1 h, 20 °C

Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 12 (1 Reaction) Steps: **1** Yield: **95%**



31-220-CAS-19951437 Steps: **1** Yield: **95%**

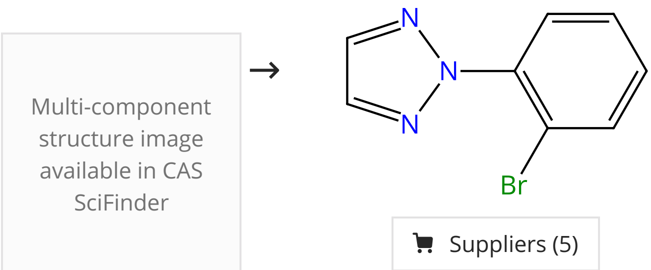
1.1 **Reagents:** *tert*-Butyl nitrite, Copper bromide ($CuBr_2$)
Solvents: Acetonitrile

Highly Selective Synthesis of 2-(2H-1,2,3-Triazol-2-yl)benzoic Acids

By: Roth, Remo; et al

Organic Process Research & Development (2019), 23(2), 234-243.

Scheme 13 (1 Reaction) Steps: **1** Yield: **95%**



Suppliers (5)

31-220-CAS-19951433 Steps: **1** Yield: **95%**

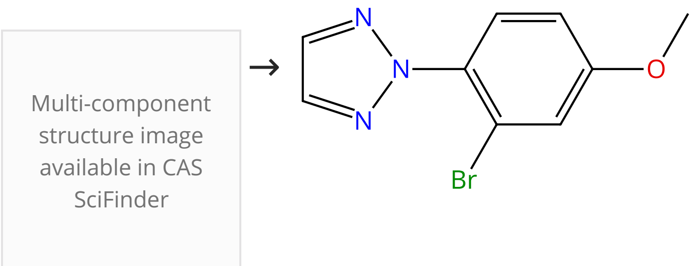
1.1 **Reagents:** *tert*-Butyl nitrite, Copper bromide ($CuBr_2$)
Solvents: Acetonitrile

Highly Selective Synthesis of 2-(2H-1,2,3-Triazol-2-yl)benzoic Acids

By: Roth, Remo; et al

Organic Process Research & Development (2019), 23(2), 234-243.

Scheme 14 (1 Reaction) Steps: **1** Yield: **95%**



31-220-CAS-19951435 Steps: **1** Yield: **95%**

1.1 **Reagents:** *tert*-Butyl nitrite, Copper bromide ($CuBr_2$)
Solvents: Acetonitrile

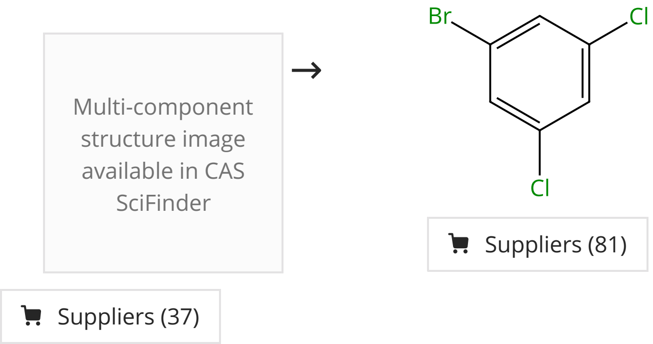
Highly Selective Synthesis of 2-(2H-1,2,3-Triazol-2-yl)benzoic Acids

By: Roth, Remo; et al

Organic Process Research & Development (2019), 23(2), 234-243.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **95%**



31-220-CAS-18799584

Steps: **1** Yield: **95%**

1.1 **Reagents:** Lithium bromide
Catalysts: Cuprous chloride, 1-Propanaminium, *N*,*N*,*N*-triethyl-, chloride (1:1) (silica supported)
Solvents: Acetonitrile; 5 min, rt

1.2 **Solvents:** Acetonitrile; 1 h, rt

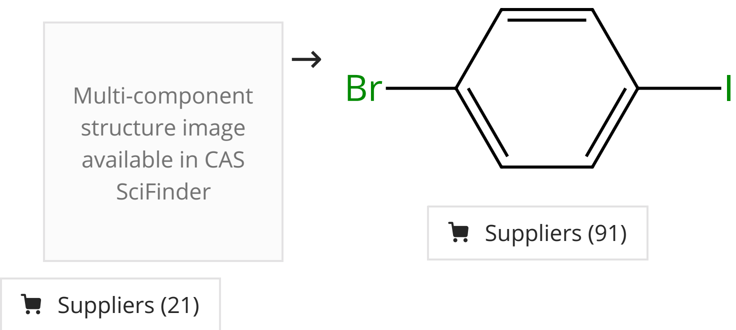
Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 16 (1 Reaction)

Steps: **1** Yield: **95%**



31-220-CAS-18799587

Steps: **1** Yield: **95%**

1.1 **Reagents:** Lithium bromide
Catalysts: Cuprous chloride, 1-Propanaminium, *N*,*N*,*N*-triethyl-, chloride (1:1) (silica supported)
Solvents: Acetonitrile; 5 min, rt

1.2 **Solvents:** Acetonitrile; 1 h, rt

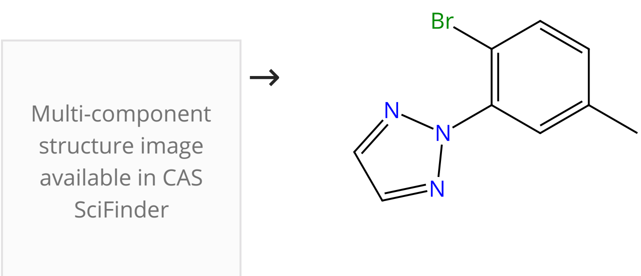
Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 17 (1 Reaction)

Steps: **1** Yield: **95%**



31-220-CAS-19951436

Steps: **1** Yield: **95%**

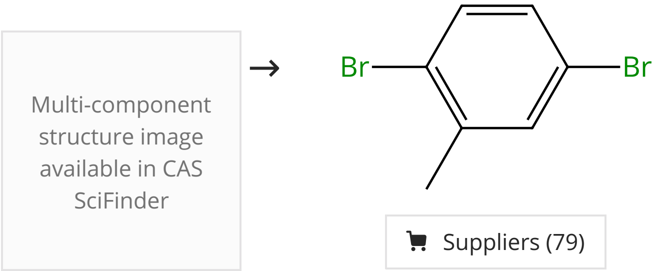
1.1 **Reagents:** *tert*-Butyl nitrite, Copper bromide ($CuBr_2$)
Solvents: Acetonitrile

Highly Selective Synthesis of 2-(2H-1,2,3-Triazol-2-yl)benzoic Acids

By: Roth, Remo; et al

Organic Process Research & Development (2019), 23(2), 234-243.

Scheme 18 (1 Reaction) Steps: **1** Yield: **94%**



Suppliers (79)

Supplier (1)

31-220-CAS-18799583 Steps: **1** Yield: **94%**

1.1 **Reagents:** Lithium bromide
Catalysts: Cuprous chloride, 1-Propanaminium, *N,N,N*-triethyl-, chloride (1:1) (silica supported)
Solvents: Acetonitrile; 5 min, rt

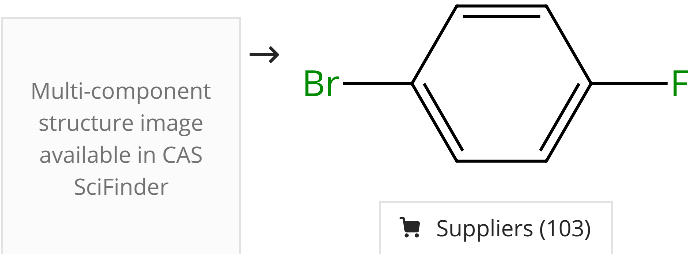
1.2 **Solvents:** Acetonitrile; 1 h, rt

Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 19 (1 Reaction) Steps: **1** Yield: **94%**



Suppliers (103)

Suppliers (30)

31-220-CAS-3735890 Steps: **1** Yield: **94%**

1.1 **Reagents:** Potassium bromide
Catalysts: Copper bromide (CuBr), Copper bromide ($CuBr_2$), Dibenzo-18-crown-6
Solvents: Acetonitrile; 2 min

1.2 **Catalysts:** *N,N,N',N'*-Tetramethylethylenediamine; 30 min, 20 °C

1.3 **Solvents:** Acetonitrile; 20 °C; 1 h, 20 °C

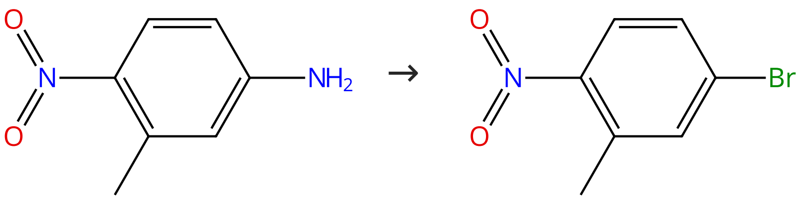
Experimental Protocols

Cu(I)/Cu(II)/TMEDA, new effective available catalyst of Sandmeyer reaction

By: Sigeev, A. S.; et al

Russian Journal of Organic Chemistry (2012), 48(8), 1055-1058.

Scheme 20 (1 Reaction) Steps: **1** Yield: **94%**



Suppliers (81)

Suppliers (63)

31-220-CAS-18799602	Steps: **1** Yield: **94%**
1.1	**Reagents:** Boron trifluoride etherate, Lithium bromide **Catalysts:** Cuprous chloride **Solvents:** Acetonitrile; 10 min, 0 °C
1.2	**Reagents:** *tert*-Butyl nitrite **Solvents:** Acetonitrile; 1 h, 20 °C

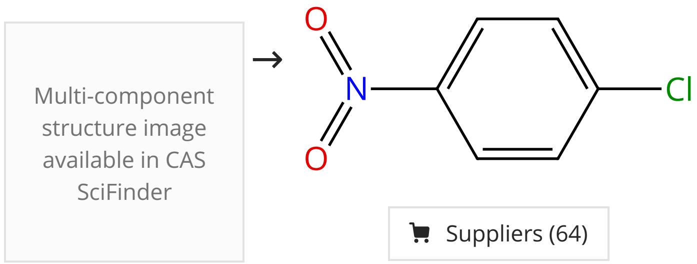
Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 21 (1 Reaction)

Steps: **1** Yield: **94%**



Suppliers (64)

Suppliers (59)

31-220-CAS-694096	Steps: **1** Yield: **94%**
1.1	**Reagents:** Potassium chloride **Catalysts:** Benzyltriethylammonium chloride, *N*,*N*,*N*′,*N*′-Tetramethylethylenediamine, Cupric chloride, Cuprous chloride **Solvents:** Acetonitrile; 30 min
1.2	**Solvents:** Acetonitrile; 20 °C; 1 h, 20 °C
Experimental Protocols	

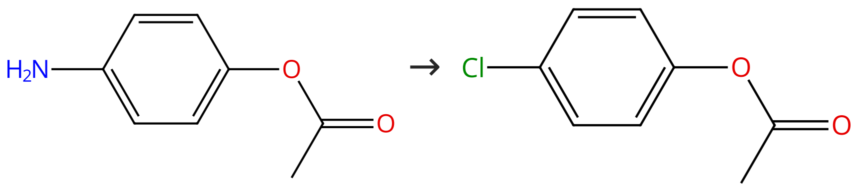
Cu(I)/Cu(II)/TMEDA, new effective available catalyst of Sandmeyer reaction

By: Sigeev, A. S.; et al

Russian Journal of Organic Chemistry (2012), 48(8), 1055-1058.

Scheme 22 (1 Reaction)

Steps: **1** Yield: **93%**



Suppliers (50)

Suppliers (68)

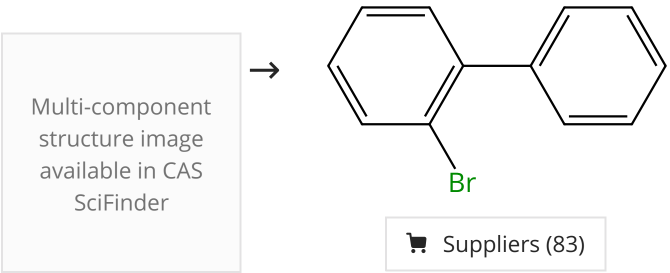
31-220-CAS-18799593	Steps: **1** Yield: **93%**
1.1	**Reagents:** Tetraethylammonium chloride, Boron trifluoride etherate **Catalysts:** Cuprous chloride, 1-Propanaminium, *N*,*N*,*N*-triethyl-, chloride (1:1) (silica supported) **Solvents:** Acetonitrile; 10 min, 0 °C
1.2	**Reagents:** *tert*-Butyl nitrite **Solvents:** Acetonitrile; 1 h, 20 °C

Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 23 (1 Reaction) Steps: **1** Yield: **93%**



Suppliers (5)

Suppliers (83)

31-614-CAS-24893068 Steps: **1** Yield: **93%**

1.1 **Reagents:** *tert*-Butyl bromide
Solvents: Methanol; -78 °C; 4 h, rt

1.2 **Solvents:** Water; rt

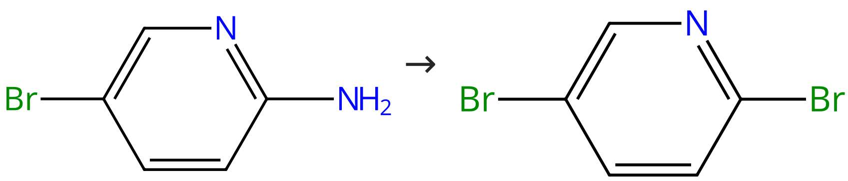
Experimental Protocols

Excitation of aryl cations as the key to catalyst-free radical arylations

By: Witzel, Sina; et al

Cell Reports Physical Science (2021), 2(2), 100325.

Scheme 24 (1 Reaction) Steps: **1** Yield: **93%**



Suppliers (99)

Suppliers (96)

31-614-CAS-24155629 Steps: **1** Yield: **93%**

1.1 **Reagents:** Bromine, Hydrogen bromide
Solvents: Water; 10 min, < 10 °C

1.2 **Reagents:** Sodium nitrite
Solvents: Water; 0 - 5 °C; 30 min, 0 - 5 °C

1.3 **Reagents:** Sodium hydroxide
Solvents: Water; < 25 °C

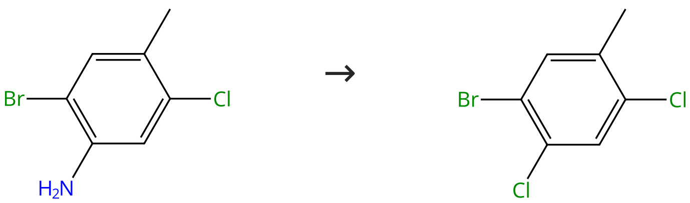
Experimental Protocols

An improved, practical, reliable and scalable synthesis of 2,5-dibromopyridine

By: Sonavane, Sachin; et al

Heterocyclic Letters (2021), 11(3), 447-452.

Scheme 25 (1 Reaction) Steps: **1** Yield: **92%**



Suppliers (64)

Suppliers (48)

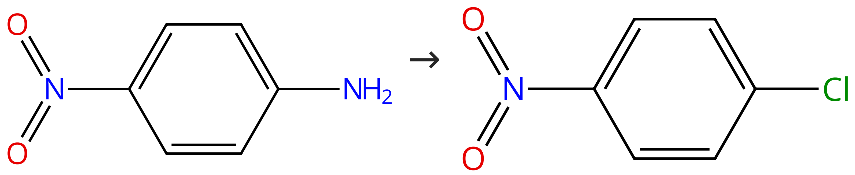
31-220-CAS-18799598	Steps: **1** Yield: **92%**
1.1	**Reagents:** Tetraethylammonium chloride, Boron trifluoride etherate **Catalysts:** Cuprous chloride, 1-Propanaminium, *N,N,N*-triethyl-, chloride (1:1) (silica supported) **Solvents:** Acetonitrile; 10 min, 0 °C
1.2	**Reagents:** *tert*-Butyl nitrite **Solvents:** Acetonitrile; 1 h, 20 °C

Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 26 (1 Reaction) Steps: **1** Yield: **91%**



Suppliers (93) Suppliers (64)

31-220-CAS-16613168	Steps: **1** Yield: **91%**
1.1	**Reagents:** *tert*-Butyl nitrite **Solvents:** Acetonitrile; 1 min, rt
1.2	**Reagents:** Cupric chloride **Solvents:** Acetonitrile, Ethylene glycol; 4 min, 82 °C

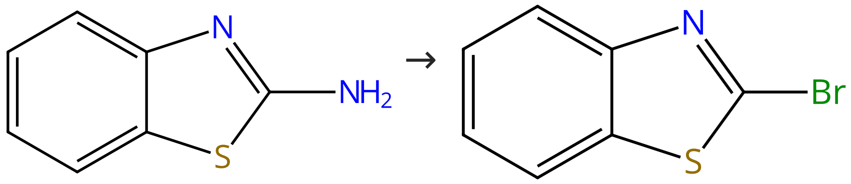
Experimental Protocols

Efficient Transposition of the Sandmeyer Reaction from Batch to Continuous Process

By: D'Attoma, Joseph; et al

Organic Process Research & Development (2017), 21(1), 44-51.

Scheme 27 (1 Reaction) Steps: **1** Yield: **91%**



Suppliers (114) Suppliers (77)

31-220-CAS-19136348	Steps: **1** Yield: **91%**
1.1	**Reagents:** *N*-Bromosuccinimide, Sodium nitrite **Solvents:** Dimethylformamide; 6 h, rt
1.2	**Reagents:** Sodium thiosulfate **Solvents:** Water; rt

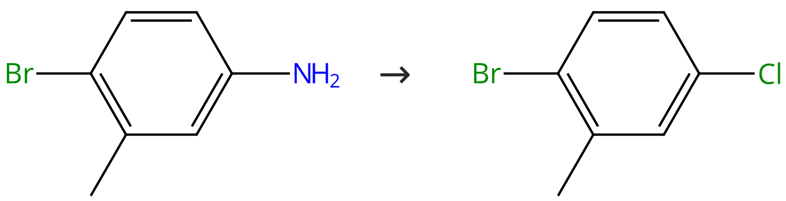
Experimental Protocols

Direct Transformation of Arylamines to Aryl Halides via Sodium Nitrite and N-Halosuccinimide

By: Mukhopadhyay, Sushobhan; et al

Chemistry - A European Journal (2018), 24(55), 14622-14626.

Scheme 28 (1 Reaction) Steps: **1** Yield: **90%**



Suppliers (81) Suppliers (65)

31-220-CAS-18799595 Steps: **1** Yield: **90%**

1.1 **Reagents:** Tetraethylammonium chloride, Boron trifluoride etherate
Catalysts: Cuprous chloride, 1-Propanaminium, *N*,*N*,*N*-triethyl-, chloride (1:1) (silica supported)
Solvents: Acetonitrile; 10 min, 0 °C

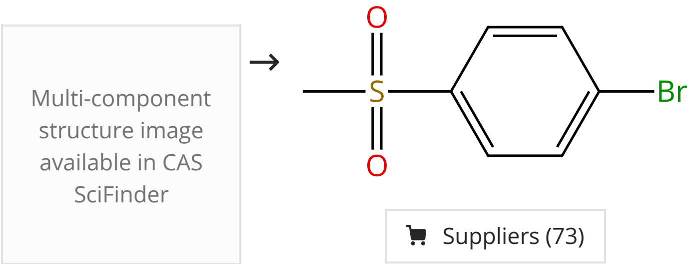
1.2 **Reagents:** *tert*-Butyl nitrite
Solvents: Acetonitrile; 1 h, 20 °C

Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 29 (1 Reaction) Steps: **1** Yield: **90%**



Supplier (1)

Suppliers (73)

31-614-CAS-46793033 Steps: **1** Yield: **90%**

1.1 **Reagents:** Acetonitrile, Bromotrichloromethane
Catalysts: Barium titanate; 1 h, rt

1.2 **Solvents:** Ethyl acetate; rt

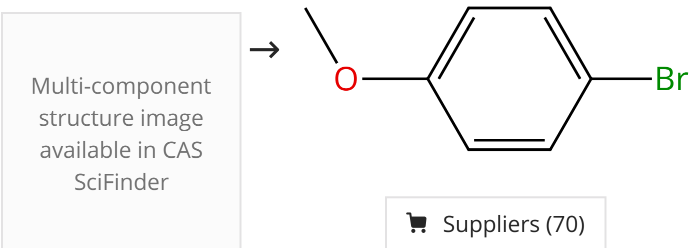
Experimental Protocols

Piezoelectric-Driven Mechanochemical Halogenation: A Transformative Pathway for Sustainable Sandmeyer Reactions via Ball Milling

By: Gao, Pan; et al

Journal of Organic Chemistry (2025), 90(31), 11223-11229.

Scheme 30 (2 Reactions) Steps: **1** Yield: **87-90%**



Suppliers (54)

Suppliers (70)

31-220-CAS-734849 Steps: **1** Yield: **90%**

1.1 **Reagents:** Potassium bromide
Catalysts: Copper bromide (CuBr), Copper bromide ($CuBr_2$), Dibenzo-18-crown-6
Solvents: Acetonitrile; 2 min

1.2 **Catalysts:** *N*,*N*,*N'*,*N'*-Tetramethylethylenediamine; 30 min, 20 °C

1.3 **Solvents:** Acetonitrile; 20 °C; 1 h, 20 °C

Experimental Protocols

Cu(I)/Cu(II)/TMEDA, new effective available catalyst of Sandmeyer reaction

By: Sigeev, A. S.; et al

Russian Journal of Organic Chemistry (2012), 48(8), 1055-1058.

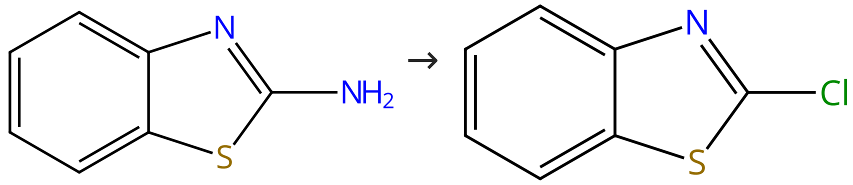
31-220-CAS-18799585	Steps: **1** Yield: **87%**
1.1	**Reagents:** Lithium bromide **Catalysts:** Cuprous chloride, 1-Propanaminium, *N,N,N*-triethyl-, chloride (1:1) (silica supported) **Solvents:** Acetonitrile; 5 min, rt
1.2	**Solvents:** Acetonitrile; 1 h, rt

Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts

By: Tarkhanova, Irina G.; et al

Mendeleev Communications (2018), 28(3), 261-263.

Scheme 31 (1 Reaction) Steps: **1** Yield: **90%**



Suppliers (114)

Suppliers (95)

31-220-CAS-19136363	Steps: **1** Yield: **90%**
1.1	**Reagents:** Chlorosuccinimide, Sodium nitrite **Solvents:** Dimethylformamide; 6 h, rt

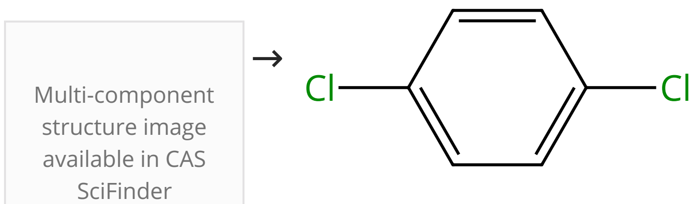
Experimental Protocols

Direct Transformation of Arylamines to Aryl Halides via Sodium Nitrite and N-Halosuccinimide

By: Mukhopadhyay, Sushobhan; et al

Chemistry - A European Journal (2018), 24(55), 14622-14626.

Scheme 32 (1 Reaction) Steps: **1** Yield: **89%**



Suppliers (107)

Suppliers (21)

31-220-CAS-409831	Steps: **1** Yield: **89%**
1.1	**Reagents:** Potassium chloride **Catalysts:** Benzyltriethylammonium chloride, *N,N,N',N'*-Tetramethylethylenediamine, Cupric chloride, Cuprous chloride **Solvents:** Acetonitrile; 30 min
1.2	**Solvents:** Acetonitrile; 20 °C; 1 h, 20 °C

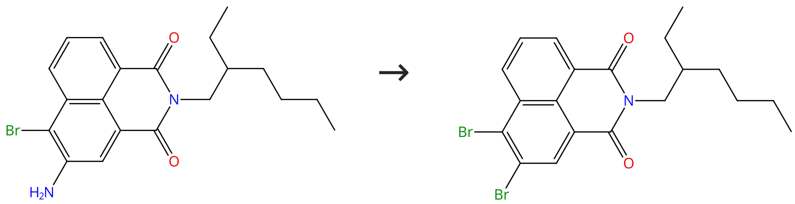
Experimental Protocols

Cu(I)/Cu(II)/TMEDA, new effective available catalyst of Sandmeyer reaction

By: Sigeev, A. S.; et al

Russian Journal of Organic Chemistry (2012), 48(8), 1055-1058.

Scheme 33 (1 Reaction) Steps: **1** Yield: **89%**



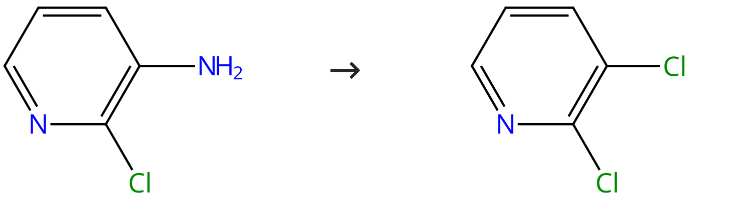
31-614-CAS-43950827	Steps: **1** Yield: **89%**
1.1	**Reagents:** Sodium nitrate, Sulfuric acid **Solvents:** Acetic acid, Acetonitrile, Water; 1 h, 0 - 5 °C; 15 min, 0 - 5 °C
1.2	**Reagents:** Copper bromide (CuBr), Hydrogen bromide **Solvents:** Water
Experimental Protocols	

A Gram Scale Synthesis of 3,4-Dihalogen Substituted 1,8-Naphthalimides

By: Anastasova, Denitsa; et al

Molbank (2024), 2024(4), M1918.

Scheme 34 (1 Reaction) Steps: **1** Yield: **89%**



Suppliers (100)   Suppliers (92)

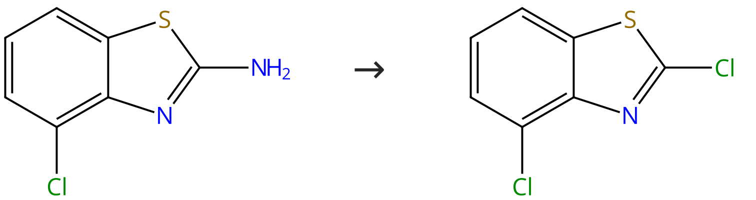
31-220-CAS-16010741	Steps: **1** Yield: **89%**
1.1	**Reagents:** Sodium nitrite, Hydrochloric acid **Solvents:** Water; rt → 5 °C; 0 - 5 °C; 1 h, 0 - 5 °C
1.2	**Reagents:** Hydrochloric acid, Cuprous chloride; 10 - 15 °C

Synthesis of 2,3-dichloropyridine

By: Liu, Wei; et al

Huaxue Shiji (2015), 37(3), 283-285, 288.

Scheme 35 (1 Reaction) Steps: **1** Yield: **89%**



Suppliers (85)   Suppliers (57)

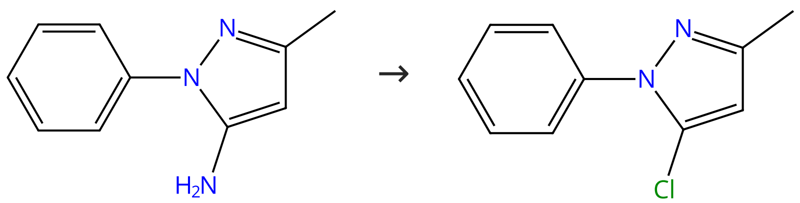
31-220-CAS-19136364	Steps: **1** Yield: **89%**
1.1	**Reagents:** Chlorosuccinimide, Sodium nitrite **Solvents:** Dimethylformamide; 6 h, rt
Experimental Protocols	

Direct Transformation of Arylamines to Aryl Halides via Sodium Nitrite and N-Halosuccinimide

By: Mukhopadhyay, Sushobhan; et al

Chemistry - A European Journal (2018), 24(55), 14622-14626.

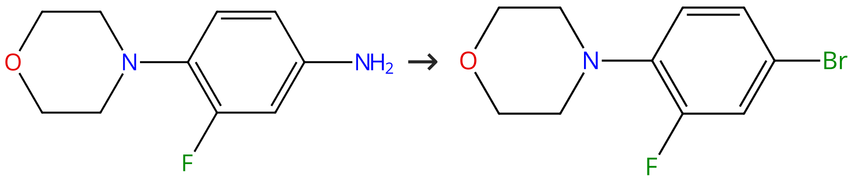
Scheme 36 (1 Reaction) Steps: **1** Yield: **88%**



Suppliers (91)   Suppliers (63)

31-220-CAS-19136362	Steps: **1** Yield: **88%**	Direct Transformation of Arylamines to Aryl Halides via Sodium Nitrite and N-Halosuccinimide
1.1	**Reagents:** Chlorosuccinimide, Sodium nitrite **Solvents:** Dimethylformamide; 6 h, rt	By: Mukhopadhyay, Sushobhan; et al
Experimental Protocols		Chemistry - A European Journal (2018), 24(55), 14622-14626.

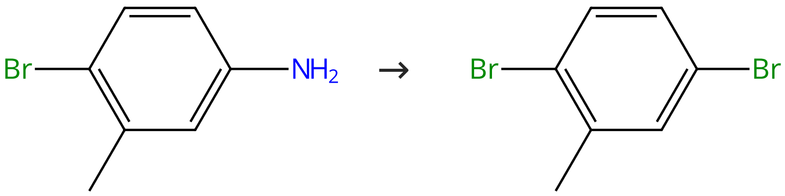
Scheme 37 (1 Reaction) Steps: **1** Yield: **88%**



Suppliers (86) Suppliers (55)

31-220-CAS-17328749	Steps: **1** Yield: **88%**	Synthesis of linezolid
1.1	**Reagents:** Sodium nitrite, Hydrogen bromide **Solvents:** Water; rt → 5 °C; 0 - 5 °C	By: Fu, Xiao-dong; et al
1.2	**Reagents:** Copper bromide (CuBr), Hydrogen bromide **Solvents:** Water; < 10 °C; 30 min, rt; rt → 50 °C; 1 h, 50 °C	Jingxi Huagong (2013), 30(8), 920-924.

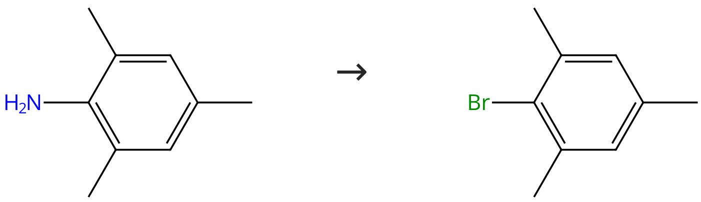
Scheme 38 (1 Reaction) Steps: **1** Yield: **88%**



Suppliers (81) Suppliers (79)

31-220-CAS-18799601	Steps: **1** Yield: **88%**	Highly efficient Sandmeyer reaction on immobilized Cu^{I}/Cu^{II}-based catalysts
1.1	**Reagents:** Boron trifluoride etherate, Lithium bromide **Catalysts:** Cuprous chloride, 1-Propanaminium, *N,N,N*-triethyl-, chloride (1:1) (silica supported) **Solvents:** Acetonitrile; 10 min, 0 °C	By: Tarkhanova, Irina G.; et al
1.2	**Reagents:** *tert*-Butyl nitrite **Solvents:** Acetonitrile; 1 h, 20 °C	Mendeleev Communications (2018), 28(3), 261-263.

Scheme 39 (1 Reaction) Steps: **1** Yield: **88%**



Suppliers (73) Suppliers (81)

31-220-CAS-19136345 Steps: **1** Yield: **88%**

1.1 **Reagents:** *N*-Bromosuccinimide, Sodium nitrite
Solvents: Dimethylformamide; 6 h, rt

1.2 **Reagents:** Sodium thiosulfate
Solvents: Water; rt

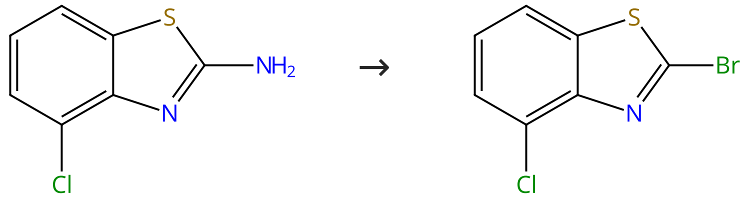
Experimental Protocols

Direct Transformation of Arylamines to Aryl Halides via Sodium Nitrite and N-Halosuccinimide

By: Mukhopadhyay, Sushobhan; et al

Chemistry - A European Journal (2018), 24(55), 14622-14626.

Scheme 40 (1 Reaction) Steps: **1** Yield: **88%**



Suppliers (85) Suppliers (53)

31-220-CAS-19136349 Steps: **1** Yield: **88%**

1.1 **Reagents:** *N*-Bromosuccinimide, Sodium nitrite
Solvents: Dimethylformamide; 6 h, rt

1.2 **Reagents:** Sodium thiosulfate
Solvents: Water; rt

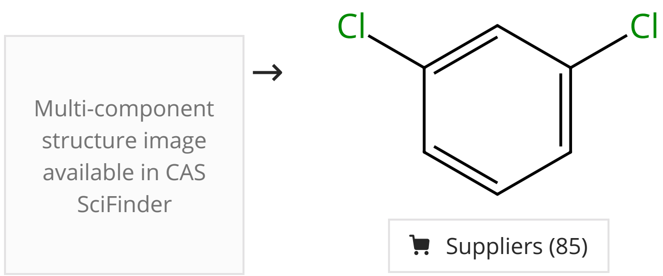
Experimental Protocols

Direct Transformation of Arylamines to Aryl Halides via Sodium Nitrite and N-Halosuccinimide

By: Mukhopadhyay, Sushobhan; et al

Chemistry - A European Journal (2018), 24(55), 14622-14626.

Scheme 41 (1 Reaction) Steps: **1** Yield: **87%**



Suppliers (85)

Suppliers (19)

31-220-CAS-2542320 Steps: **1** Yield: **87%**

1.1 **Reagents:** Potassium chloride
Catalysts: Benzyltriethylammonium chloride, *N*,*N*,*N*′,*N*′-Tetramethylethylenediamine, Cupric chloride, Cuprous chloride
Solvents: Acetonitrile; 30 min

1.2 **Solvents:** Acetonitrile; 20 °C; 1 h, 20 °C

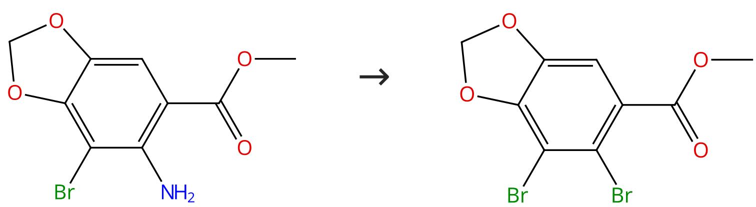
Experimental Protocols

Cu(I)/Cu(II)/TMEDA, new effective available catalyst of Sandmeyer reaction

By: Sigeev, A. S.; et al

Russian Journal of Organic Chemistry (2012), 48(8), 1055-1058.

Scheme 42 (1 Reaction) Steps: **1** Yield: **87%**



Suppliers (4) Supplier (1)

31-220-CAS-7540318 Steps: **1** Yield: **87%**

1.1 **Reagents:** Sodium nitrite, Hydrogen bromide
Solvents: Water; rt → 5 °C; 0 - 5 °C; 20 min, 0 - 5 °C

1.2 **Reagents:** Copper bromide (CuBr), Hydrogen bromide
Solvents: Water; 30 min, 0 - 5 °C; 20 min, 5 °C → 80 °C

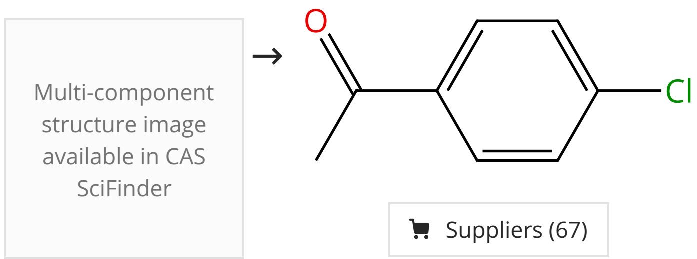
1.3 **Reagents:** Water; overnight, rt

Synthesis of bromination derivatives of 1,3-benzodioxole

By: Feng, Xiue; et al

Xiandai Huagong (2010), 30(11), 61-63.

Scheme 43 (1 Reaction) Steps: **1** Yield: **87%**



Suppliers (67)

Suppliers (5)

31-220-CAS-4952228 Steps: **1** Yield: **87%**

1.1 **Reagents:** Potassium chloride
Catalysts: Benzyltriethylammonium chloride, *N,N,N′,N′*-Tetramethylethylenediamine, Cupric chloride, Cuprous chloride
Solvents: Acetonitrile; 30 min

1.2 **Solvents:** Acetonitrile; 20 °C; 1 h, 20 °C

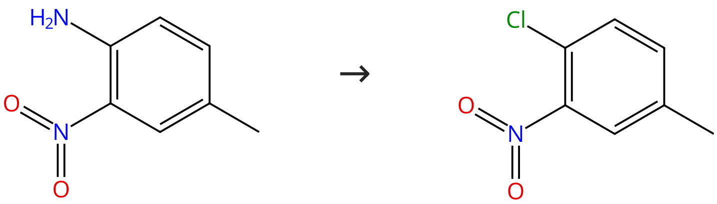
Experimental Protocols

Cu(I)/Cu(II)/TMEDA, new effective available catalyst of Sandmeyer reaction

By: Sigeev, A. S.; et al

Russian Journal of Organic Chemistry (2012), 48(8), 1055-1058.

Scheme 44 (1 Reaction) Steps: **1** Yield: **87%**



Suppliers (73)

Suppliers (72)

31-220-CAS-16609790 Steps: **1** Yield: **87%**

1.1 **Reagents:** *tert*-Butyl nitrite
Solvents: Acetonitrile; 1 min, rt

1.2 **Reagents:** Cupric chloride
Solvents: Acetonitrile, Ethylene glycol; 4 min, 82 °C

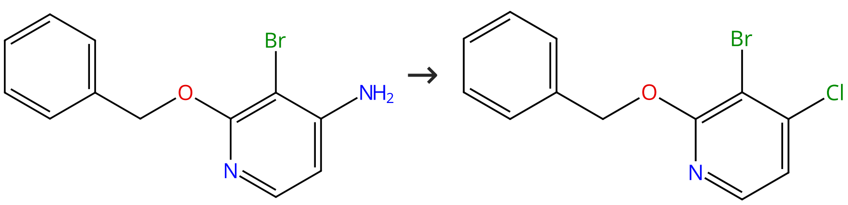
Experimental Protocols

Efficient Transposition of the Sandmeyer Reaction from Batch to Continuous Process

By: D'Attoma, Joseph; et al

Organic Process Research & Development (2017), 21(1), 44-51.

Scheme 45 (1 Reaction) Steps: **1** Yield: **86%**



Suppliers (30)

Suppliers (9)

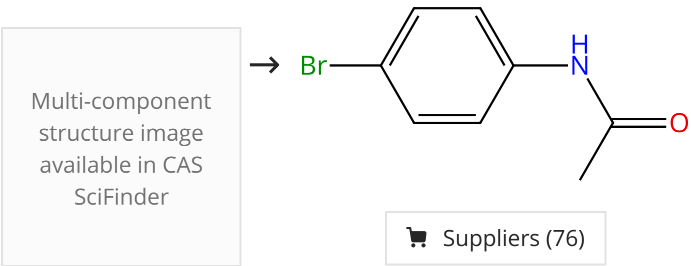
31-220-CAS-15480566	Steps: **1** Yield: **86%**
1.1	**Reagents:** *tert*-Butyl nitrite, Cupric chloride **Solvents:** Acetonitrile; rt; 15 min, rt; rt → 0 °C
1.2	**Solvents:** Acetonitrile; 0 °C; 1 h, 0 °C; 16 h, 0 °C → rt
Experimental Protocols	

Concise Entries to 4-Halo-2-pyridones and 3-Bromo-4-halo-2-pyridones

By: Honraedt, Aurelien; et al

Synlett (2016), 27(1), 67-69.

Scheme 46 (1 Reaction) Steps: **1** Yield: **86%**



Suppliers (76)

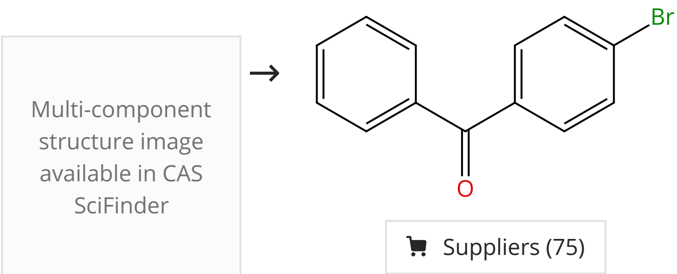
31-614-CAS-46793035	Steps: **1** Yield: **86%**
1.1	**Reagents:** Acetonitrile, Bromotrichloromethane **Catalysts:** Barium titanate; 1 h, rt
1.2	**Solvents:** Ethyl acetate; rt
Experimental Protocols	

Piezoelectric-Driven Mechanochemical Halogenation: A Transformative Pathway for Sustainable Sandmeyer Reactions via Ball Milling

By: Gao, Pan; et al

Journal of Organic Chemistry (2025), 90(31), 11223-11229.

Scheme 47 (1 Reaction) Steps: **1** Yield: **86%**



Suppliers (75)

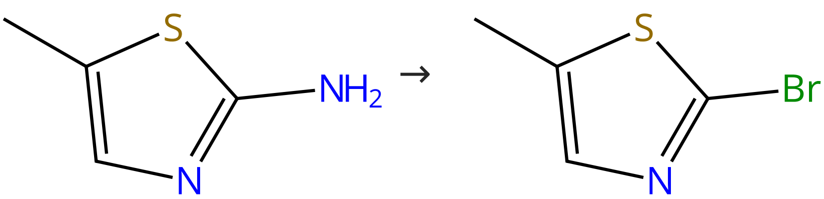
31-614-CAS-37098539	Steps: **1** Yield: **86%**
1.1	**Reagents:** Sodium bromide **Catalysts:** *N*,*N*-Bis[(diphenylphosphino)methyl]-3-(triethoxysilyl)-1-propanamine (bis(diphenylphosphinomethyl)amino-modified MCM-41-immobilized gold com...) **Solvents:** Acetonitrile; 5 h, 50 °C
Experimental Protocols	

Heterogeneous gold-catalyzed Sandmeyer coupling of aryldiazonium salts with sodium bromide or thiols for constr ucting C-Br and C-S bonds

By: Li, Jiajia; et al

Dalton Transactions (2023), 52(29), 10045-10057.

Scheme 48 (1 Reaction) Steps: **1** Yield: **86%**



Suppliers (98) Suppliers (83)

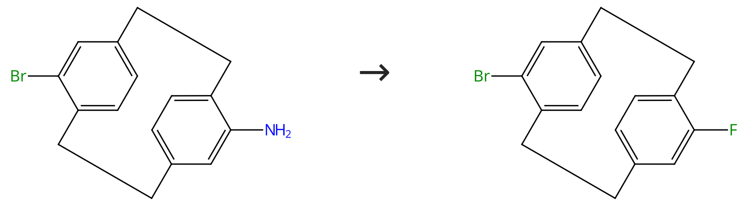
31-220-CAS-19136351	Steps: **1** Yield: **86%**
1.1	**Reagents:** *N*-Bromosuccinimide, Sodium nitrite **Solvents:** Dimethylformamide; 6 h, rt
1.2	**Reagents:** Sodium thiosulfate **Solvents:** Water; rt
Experimental Protocols	

Direct Transformation of Arylamines to Aryl Halides via Sodium Nitrite and N-Halosuccinimide

By: Mukhopadhyay, Sushobhan; et al

Chemistry - A European Journal (2018), 24(55), 14622-14626.

Scheme 49 (1 Reaction) Steps: **1** Yield: **85%**



31-220-CAS-7380747	Steps: **1** Yield: **85%**
1.1	**Reagents:** Tetrafluoroboric acid **Solvents:** Ethanol; rt → 60 °C; 10 min, 60 °C; 60 °C → - 5 °C
1.2	**Reagents:** Isoamyl nitrite; -5 °C; 1 h, - 5 °C
1.3	**Solvents:** Toluene; rt → 60 °C
Experimental Protocols	

Planar Chirality Change in Dediazoniation Reactions of Paracyclophanes and Mechanistic Implication

By: He, Fuyan; et al

Organic Letters (2012), 14(21), 5436-5439.